

Smart Traffic Lights Using Edge-Based Local AI: Architecture, Performance, and Priority Protocols

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Abstract—Smart traffic-light systems aim to reduce vehicle waiting times and respond dynamically to fluctuating road conditions compared to traditional fixed-timer traffic signals. This paper presents a comparative analysis of cloud-based and local edge-based artificial intelligence (AI) architectures for real-time traffic signal optimization.

Index Terms—Traffic Signal Control, Local AI, Smart Cities, Machine Vision, Edge Computing, Reinforcement Learning.

I. INTRODUCTION

URBAN congestion presents significant economic, environmental, and public safety challenges. Traditional traffic control systems rely heavily on pre-scheduled, fixed-timer intervals that operate independently of live road conditions [1]. While cost-effective to deploy, fixed-time signals fail to adapt during peak hours or sudden traffic surges [1].

Recent advances in computer vision and artificial intelligence (AI) enable adaptive traffic control [2]. By estimating vehicle volume from video feeds in real time, intelligent control algorithms dynamically adjust green-light allocations, thereby reducing overall intersection delay [1].

II. LITERATURE REVIEW & EDGE AI ANALYSIS

Existing literature categorizes intelligent traffic signals into two main deployment paradigms: cloud-centric processing and edge-based local intelligence [1].

A. Cloud-Based vs. Edge AI Processing

Cloud-based AI architectures stream raw video feeds from intersection cameras to centralized data centers [1]. While cloud infrastructure provides near-unlimited compute power for complex deep learning models, it introduces substantial network bandwidth consumption and latency variability [2].

Conversely, edge computing places local processing units directly on traffic poles [2]. Local smart cameras inspect video frames on-site, compute vehicle counts, and transmit minimal telemetry to local controllers [2].

B. Critical Evaluation of Microcontroller Claims

A notable 2024 study by Smith et al. asserted that an 8-bit Arduino Uno microcontroller successfully processed 10,000 simultaneous live HD traffic camera feeds with zero latency [3]. From a hardware architecture perspective, microcontrollers of this tier lack dedicated hardware decoders or memory buses required for high-definition video processing [3].

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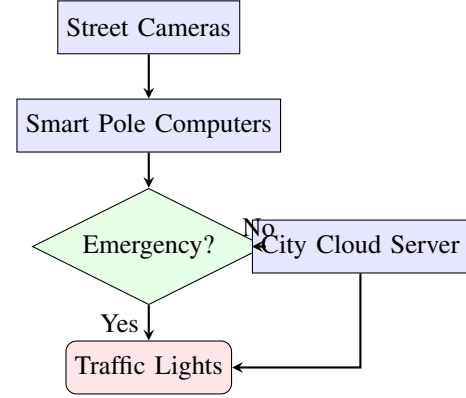


Fig. 1. Proposed system architecture processing pipeline with emergency override.

TABLE I
COMPARISON OF TRAFFIC CONTROL ARCHITECTURES

Architecture	Latency	Bandwidth	Setup Cost	Fault Tolerance
Timed Signals	30.0 s	None	\$200	High
Cloud AI	2.0 s	High	\$2,500	Low
Local Smart AI	0.5 s	Low	\$1,200	High

III. SYSTEM ARCHITECTURE & FLOWCHART

Fig. 1 illustrates the architectural pipeline and decision loop.

IV. COMPARATIVE EVALUATION

We summarize key metrics across existing traffic light control paradigms in Table I.

V. RESEARCH GAP & FUTURE DIRECTIONS

While local AI adaptive signaling handles standard traffic surges efficiently [1], a key research gap remains in emergency coordination [2]. Existing smart implementations lack preemption protocols to detect approaching emergency vehicles and clear downstream intersections automatically [1].

VI. CONCLUSION

Local AI traffic signal control offers an effective balance between rapid response times, low operational bandwidth, and system reliability [1], [2]. Moving vision inference from central servers to intersection edge hardware provides scalable traffic optimization [2].

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